Project Proposal for Senior Design Project AI-Powered Supply Chain & Inventory Intelligence System

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1. Introduction:

Small and Medium Enterprises (SMEs), especially small retail stores and supermarkets, play a vital role in the economy. Despite their importance, most SMEs still rely on manual methods such as notebooks or basic spreadsheets to manage inventory and supplier decisions. These methods are error-prone, time-consuming, and reactive in nature.

Inventory mismanagement often leads to frequent stockouts, overstocking, poor cash flow, and loss of customer trust. While large enterprises use advanced ERP systems, such solutions are expensive and complex, making them unsuitable for small businesses.

This project proposes an **AI-assisted, low-cost inventory and supplier decision support system** designed specifically for small supermarkets. The system uses automation workflows and simple AI reasoning to predict stockouts, analyse risks, and recommend reorder actions, helping store owners make informed decisions in advance.

2. PROBLEM STATEMENT:

- Small retail stores face significant challenges in managing inventory efficiently due to the absence of predictive and decision-support tools. Inventory decisions are usually made based on intuition rather than data, leading to frequent stock shortages or excess inventory. Supplier selection is also manual, relying on memory instead of performance metrics such as cost, delivery time, and reliability.

- Existing inventory management solutions are either too basic or too expensive for SMEs. There is a need for an affordable, easy-to-use system that assists store owners by predicting inventory risks and recommending appropriate actions before problems occur.

3. OBJECTIVES:

The main objectives of this project are:

1. To design a simple inventory management system suitable for small supermarkets.
2. To predict product stockout dates based on current stock and sales patterns.
3. To recommend reorder quantities to avoid stock shortages.
4. To compare suppliers based on cost, delivery time, and reliability.
5. To analyse inventory-related risks and present them clearly to the user.
6. To reduce manual effort and improve decision-making using automation and AI assistance.

4. Scope of the Project:

The scope of the project includes:

- Managing product inventory data such as stock levels and daily sales.
- Predicting stock depletion timelines using logical rules and AI support.
- Providing reorder and supplier recommendations.
- Displaying insights through a simple web dashboard.
- Supporting a single store (small supermarket) use case.

The project does not include payment processing, real-time POS integration, or live supplier website scraping. The focus is on decision support rather than full automation.

5. Literature Survey:

Inventory management and demand forecasting have been widely studied areas in supply chain management research. Traditional inventory management systems primarily focus on tracking stock levels and triggering reorder alerts when inventory falls below a predefined threshold. These systems are largely reactive in nature and depend heavily on historical sales data, which limits their ability to respond to dynamic market conditions.

Several studies have explored demand forecasting techniques using statistical and machine learning models. While these approaches improve forecast accuracy, most of them rely only on past sales data and require extensive model training and historical datasets. Such systems are typically implemented in large enterprises and are not feasible for small and medium enterprises (SMEs) due to high implementation costs, technical complexity, and maintenance requirements.

Recent research has highlighted the effectiveness of multi-factor demand forecasting, where additional parameters such as weather conditions, local events, and market trends are incorporated along with sales history. However, these multi-factor approaches are predominantly found in enterprise-level systems and academic prototypes. There is very limited evidence of cost-effective, SME-oriented implementations that integrate these factors into a single operational system.

Supplier selection and evaluation have also been studied extensively, with existing research focusing on cost optimization, delivery performance, and reliability metrics. Although these models provide structured methods for supplier comparison, they are often implemented as standalone decision models or embedded within expensive enterprise resource planning (ERP) systems. SMEs typically lack access to such tools and continue to rely on manual supplier selection based on experience and intuition.

Another emerging area of interest is the use of low-code and no-code platforms for business process automation. While these platforms are increasingly adopted in industry for workflow automation, academic literature shows a lack of comprehensive studies demonstrating their application in end-to-end supply chain decision support systems, particularly for SMEs.

Most existing studies treat forecasting, supplier evaluation, and purchase decision processes as separate problems rather than as an integrated workflow.

Furthermore, the use of pre-trained large language models (LLMs) for demand forecasting and decision support is still under-explored in academic research. Existing works primarily focus on traditional machine learning models that require training and tuning, whereas the potential of zero-training AI-assisted reasoning for inventory decision support remains largely unexplored.

Based on the literature review, it is observed that while individual components such as demand forecasting, supplier selection, and inventory control have been studied independently, there is a clear research gap in developing an affordable, integrated, and SME-focused system that combines these components using workflow automation and AI assistance. This project aims to address this gap by designing and implementing a practical, low-code inventory and supplier decision support system tailored for small retail environments.

6. Proposed Methodology:

The proposed system follows a modular workflow-based architecture:

I. **Data Input: -**
The store owner (admin) manually enters product details, current stock, daily sales, and supplier information through a web interface.

II. **Inventory Processing: -**
Automated workflows calculate remaining stock and estimate the number of days before stock depletion.

III. **AI-Assisted Prediction: -**
AI logic is used to explain stockout predictions and suggest reorder quantities in an understandable manner.

IV. **Supplier Evaluation: -**
Suppliers are compared based on predefined attributes such as price, delivery time, and reliability score.

V. **Risk Analysis: -**
Products are classified into risk categories such as high risk of stockout or overstock.

VI. **Dashboard Visualization: -**
All results are presented on a web dashboard with clear indicators for easy decision-making.

7. Hardware Requirements:

- Laptop or Desktop Computer
- Minimum 8 GB RAM
- Stable Internet Connection

8. Software Requirements:

- Web Browser (Google Chrome or equivalent)
- n8n (workflow automation platform)
- Airtable or Supabase (database)
- No-code / low-code frontend builder
- AI API (free-tier usage)
- Operating System: Windows / Linux / macOS

9. Expected Outcomes:

- A working prototype of an inventory decision support system.
- Accurate prediction of product stockout timelines.
- Reduced manual effort in inventory monitoring.
- Clear visualization of inventory risks and reorder suggestions.
- Improved supplier selection decisions for small retailers.

10. Project Schedule:

Phase	Duration (Weeks)	Activities	Review Alignment
Phase 1	**Week 1**	**Finalization of problem statement, objectives, and system scope**	**—**
Phase 2	**Week 2**	**Initial n8n setup, database connection, basic inventory workflows**	**Review 1**
Phase 3	**Weeks 3–4**	**Database schema design, product and supplier management workflows**	**—**

Phase 4	**Weeks 5–6**	**Core inventory logic and stockout calculation workflows**	**—**
Phase 5	**Weeks 7–8**	**Dashboard data aggregation and frontend integration**	**—**
Phase 6	**Week 9**	**Partial system integration and demo preparation**	**Review 2**
Phase 7	**Weeks 10–11**	**AI-assisted prediction and reorder recommendation workflows**	**—**
Phase 8	**Weeks 12–13**	**Risk analysis and compliance indicators**	**—**
Phase 9	**Weeks 14–15**	**System testing, validation, and demo data preparation**	**—**
Phase 10	**Week 16**	**Final documentation, PPT preparation, and viva readiness**	**Final Review**

11. Applications:

- Small supermarkets
- Kirana stores
- Local retail shops
- Mini-marts and convenience stores
- Small inventory-based businesses

12. Future Scope:

- Integration with POS systems for automatic sales updates.
- Inclusion of real-time supplier data and live price tracking.
- Advanced demand forecasting using external factors such as weather and events.
- Mobile application support.
- Multi-store inventory management.

13. Conclusion:

This project presents a practical and affordable solution to inventory management challenges faced by small retail stores. By combining workflow automation with AI-assisted reasoning, the system shifts inventory management from a reactive to a proactive approach. The proposed solution is easy to use, cost-effective, and suitable for real-world SME environments, making it a strong candidate for an engineering final-year project.

14. References:

1. Research articles on low-code automation and SME inventory management
2. Online documentation of n8n workflow automation platform